

Animesh Karmakar

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SUMMARY

AI & Data Science graduate with practical experience in machine learning model development, data-driven analysis, and Generative AI applications. Strong interest in Large Language Models, RAG, and prompt engineering, with a drive to build and deploy production-ready AI solutions.

SKILLS

- **Languages & Databases:** Python, SQL, C++, MySQL, MongoDB
- **Machine Learning:** NumPy, Pandas, Scikit-Learn, Matplotlib, PyTorch, TensorFlow, OpenCV, EDA
- **Generative AI:** RAG, LangChain, Vector Databases (Qdrant, ChromaDB), Prompt Engineering
- **Tools & Cloud:** Docker, AWS (EC2, S3, RDS), Git/GitHub, CI/CD
- **Core CS:** OOP, Computer Networks, Operating Systems, DBMS

PROJECTS

LegalDocAI – Generative AI System for Legal Document Analysis

GitHub

Python, LangChain, RAG, OCR, FastAPI, PostgreSQL, Streamlit, Gemini API, Sentence-Transformers, Docker

- Engineered an OCR ingestion pipeline using **LangChain Recursive Chunking** and **SHA-256 cryptographic hashing** to intercept duplicate PDFs, completely eliminating redundant **HuggingFace sentence embedding inference costs**.
- Implemented a high-accuracy **Hybrid Search architecture (BM25 + Cosine Vector Similarity)** that optimized the classification of unstructured contract clauses and specific risk-scoring accuracy.
- Built a full-stack **RAG pipeline using LangChain, Google Gemini, and LLaMA-3**, deploying a real-time contract chatbot that generates deterministic, citation-backed executive risk summaries.

TradeGuard AI – Multi-Agent System for Global Trade & Customs Compliance

GitHub

Python, LangGraph, Qdrant, LangChain, FastAPI, Docker, BGE Embeddings, Streamlit, Gemini API

- Developed an AI agent platform for classifying products into official **HS tariff codes**, utilizing a **LangGraph Supervisor** node to parse requests into shared state and route queries dynamically to specialized worker agents.
- Engineered a hybrid search engine in **Qdrant** combining **BGE Dense Embeddings and Sparse BM25Plus via RRF Fusion**, instantly matching colloquial product descriptions to strict tariff taxonomies with maximum accuracy.
- Programmed a production-hardened verification gate using **LangChain** that parses structured model outputs and audits generated claims against live regulatory source text to block hallucinated tariff rates.

Sangrahaak – Predictive Demand Forecasting & Inventory AI/ML Platform

GitHub

Python, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, ARIMA, Flask, MongoDB, Docker, AWS EC2

- Engineered a full-stack, multi-depot inventory platform handling **100+ SKUs**, integrating automated low-stock alerting and low-latency **WebSocket pipelines** for real-time visualization.
- Improved demand forecasting precision to a **10.11% MAPE** and achieved a **0.91 AUC-ROC** on imbalanced classes by training and optimizing **XGBoost** and **ARIMA** models.
- Developed and containerised a **RAG-based conversational AI** using **LangChain** and **LLaMA-3.2-70B**, deploying the entire system end-to-end via **Docker** on **AWS EC2** using **REST APIs**.

EDUCATION

Terna Engineering College, Mumbai University

2022 – 2026

B.E. in Artificial Intelligence & Data Science

CGPA: 7.5 / 10

R.K. Talreja College, Ulhasnagar

2022

Higher Secondary Certificate (HSC)

79.17%

Sai English High School, Kalyan

2020

Secondary School Certificate (SSC)

90.20%

HACKATHONS & ACHIEVEMENTS

- **Gen-AI Hackathon 2025:** Selected in Top 75 teams nationally out of 418+; engineered a healthcare analytics pipeline with automated feature extraction and structured data transformations under real-world time constraints.
- **MumbaiHacks 2025:** Developed a real-time misinformation detection platform; engineered a streaming NLP pipeline to ingest data, cross-reference structured facts, and compute statistical confidence scores at scale.

CERTIFICATIONS

- **Generative AI Fundamentals** – AI Fundamentals, Data-Driven Applications, Prompt Engineering (Link)
- **MySQL Certification** – Advanced SQL, Querying, Joins, Window Functions, Aggregation, Subqueries (Link)
- **AWS Cloud Certification** – Cloud Computing, EC2, S3, Data Storage, Cloud Data Infrastructure (Link)